An La

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Education

2021 - present MS/PhD (Computer Science) at **University of Massachusetts Amherst** (GPA: 3.82/4.0)
Advisor: Professor Hung Le
Anticipated graduation: Sep 2027

2013 - 2017 Bachelor's Degree at **Honors Program, VNU-HCMUS, Vietnam** (GPA: 3.57/4.0)
Information Technology, Graduated with distinction.
Final Year Thesis: *Deep Neural Network for Vietnamese News Recognition* (Grade: 10.0/10.0)

Publications

La, A., Le, H., Solomon, S., Than, C., Vinayak, Yang, S., & Zhang, C. (2025, August) *Optimal Bounds for Spanners and Tree Covers in Doubling Metrics*. In SoCG'26: The 42th International Symposium on Computational Geometry.

La, A., & Le, H. (2025, June). *Dynamic Locality Sensitive Orderings in Doubling Metrics*. In STOC'25: Proceedings of the 57th Annual ACM Symposium on Theory of Computing (pp. 1178–1189).

La, A., & Le, H. (2024, August). *New weighted additive spanners*.

La, A., Vo, P., & Vu, T. (2019, July). *Adaptive Collaborative Filtering for Recommender System*. In International Conference on Conceptual Structures (pp. 117-130).

La, A. N. T.*, Nguyen, D. P.*, Pham, N. M., & Vu, Q. H. (2018). *Multi-modal video retrieval using Dilated Pyramidal Residual network*. Science and Technology Development Journal-Natural Sciences, 2(5), 138-143. [1]

Bumb, M., Vemulapalli, A., Jella, S.H.V.P., Gupta, A., **La, A.**, Rossi, R.A., Chen, H., Dernoncourt, F., Ahmed, N.K. and Wang, Y., (2025, July). *Forecasting Time Series with LLMs via Patch-Based Prompting and Decomposition*, to be appeared in PACLIC 39.

Experiences

Oct 2026 - Jan 2027: Applied Scientist Intern at Amazon

- ⋆ Design a forecasting system for lifetime demand of Amazon devices.
- ⋆ Technical skills: *quantitative analysis*, *forecasting design*, *machine learning and statistics*.

2021 - now: Research and Teaching Assistant at Theory CS Group/UMass Amherst

- ⋆ Study **dynamic** data structures and algorithms (spatial-indexing structures) for computational geometry applications, e.g., nearest-neighbor search, distance preservers, k-center clustering.
- ⋆ Study online algorithms with machine learning predictions.
- ⋆ Teaching Assistant: Algorithms for Data Science, Advanced Algorithms.
- ⋆ Technical skills: *theoretical analysis*, *dynamic-algorithm design*, *data structure design*.

Sep - Dec, 2025: Applied Scientist Intern at Amazon

- ⋆ Studied **knowledge-based** system with Generative AI for time-series forecasting problems.
- ⋆ Designed an exploratory AI assistant for data scientists to analyze time-series data with statistical models.
- ⋆ Technical skills: *forecasting design*, *prompt engineering*, *LLMs applications*.

July - August, 2025: Data Scientist Intern at GeoComply

- ⋆ Designed and implemented a dynamic clustering framework for robust classification problems.
- ⋆ Designed and developed a **knowledge-based** multi-agent system for fraud detection.
- ⋆ Improved time efficiency by 4x and discovered new fraud patterns.
- ⋆ Technical skills: *data structure design*, *agentic-system design*, *LLMs applications*.

2020 - 2021: Data Scientist at PrimeData, Vietnam github/anla11/analytic_marketing

- ⋆ Designed and implemented an automatic **knowledge-based** framework for algorithmic marketing applications.
- ⋆ Technical skills: *quantitative analysis*, *Bayesian machine learning* and *probabilistic programming*.

[1] *These authors contributed equally to the work.

2017 - 2019: Data Scientist at FPT Telecom, FPT Group, Vietnam github/anla11/adaptive_cf_recsys

- ⋆ Designed and implemented a graph-based model with multiple evaluation metrics for a recommender system.
- ⋆ Increased precision by 6% while maintaining diversity, coverage, and congestion.
- ⋆ Technical skills: *content-based, collaborative-filtering, and graph-based techniques.*

2016 - 2017: Implement image-processing algorithms with standard techniques using OpenCV

- ⋆ Style extraction using GIST and LAB features: github/anla11/style-extraction-for-images
- ⋆ Text region extraction with morphology-based methods: github/anla11/text-region-extract
- ⋆ Image enhancement: github/anla11/image-enhancement

Skills

Theory	Design data structures & algorithms, quantitative analysis and modeling.
Programming	5+ years of implementing projects with Python: implement machine learning models (Tensor Flow, PyTorch, Scikit-Learn); process, analyze and visualize data (Numpy, Pandas, Seaborn). Proficient in C++ to implement algorithms in competitive programming contests and Image Processing (OpenCV).

Academic Activities

June. 2026	Attended Computational Geometry 2026 (SoCG'26) Presented *Optimal Bounds for Spanners and Tree Covers in Doubling Metrics*
June. 2025	Attended Annual ACM Symposium on Theory of Computing (STOC'25) Presented *Dynamic LSO in Doubling Metrics*
Spring 2025	Industry Mentorship Independent Study - Data Science. Supervised Master students' project
Spring 2025	Early Research Scholar Programs. Supervised undergraduate research projects
Jan. 2025	ACM-SIAM Symposium on Discrete Algorithms (SODA25)
June. 2024	DIMACS Tutorial on Fine-graned Complexity
Jan. 2024	ACM-SIAM Symposium on Discrete Algorithms (SODA24)
Nov. 2022	IEEE 63rd Annual Symposium on Foundations of Computer Science (FOCS22)
Aug. 2022	FODSI Sublinear algorithms summer school and workshop
June. 2019	Attended 24th International Conference on Conceptual Structures Presented *Adaptive Collaborative Filtering for Recommender Systems*
Aug. 2017	Attended the 3rd Workshop on Statistical Modeling and Applications at VNU-HCMUS Topic: Bayesian Models Inference and Statistical Decision Making

Selective Taken Courses

Distributed and Operating Systems, Algorithms with Predictions, Randomized Algorithms, Algorithms for Data Science, and Advanced algorithms at UMASS Amherst
Bayesian Methods for Machine Learning - National Research University Higher School of Economics
Probabilistic Graphical Models 1: Representation - Stanford University
Bayesian Statistics: Techniques and Models - University of California, Santa Cruz
Bayesian Statistics: From Concept to Data Analysis - University of California, Santa Cruz

Honors and Awards

Dec. 2016	National Vietnam award for Outstanding Female Students in Science and Technology
Aug. 2016	Awards from Facebook Hackathon Vietnam 2016 *1st prize* of Most Innovative Product *2nd prize* of Best Product in Facebook Marketing Category
2012 - 2014	Vallet Scholarship (South Region) for Excellent Students - https://rvn-vallet.org/
2014	*2nd prize* in ACM-ICPC Vietnam National 1st Round
2013	*3rd prize* in Informatics at the Vietnam National Excellent Student Exam
2012	*Honourable Mention* in Informatics at the National Excellent Student Exam
2011	*Silver Medal* in Informatics at The Traditional 30/4 Olympic Competition